COSC 39: Theory of Computation

Problem 1.
Imagine we want to describe a deck of Poker cards using regular expressions. Here a Poker deck may have card repetition and more than 52 cards. We assume that *there are no Jokers*; every card is of one of the four suits: spades $\spadesuit$, hearts $\heartsuit$, diamonds $\diamondsuit$, and clubs $\clubsuit$. We say that spades and clubs are of the color *black*, while hearts and diamonds are *red*.
Design a regular expression for the following collection of decks, as a language over alphabet $\Sigma = \{\spadesuit, \heartsuit, \diamondsuit, \clubsuit\}$:

- The collection of all Poker decks where no three cards of the same color occur consecutively.

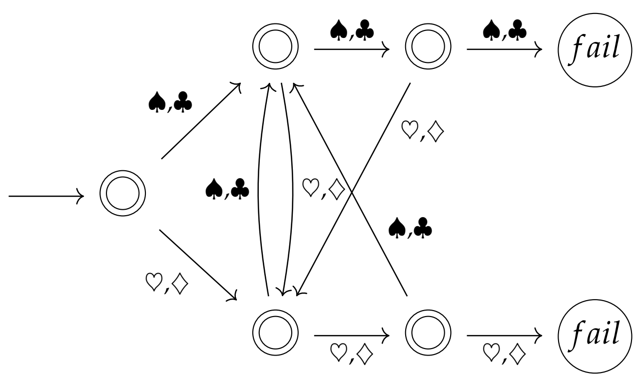
(No formal proofs required, but explanation of your construction in a few English sentences would be helpful.)

Solution.

$$(\varepsilon + (\heartsuit + \diamondsuit) + (\heartsuit + \diamondsuit)^2)(((\spadesuit + \clubsuit) + (\spadesuit + \clubsuit)^2)((\heartsuit + \diamondsuit) + (\heartsuit + \diamondsuit)^2))^\star(\varepsilon + (\spadesuit + \clubsuit) + (\spadesuit + \clubsuit)^2)$$

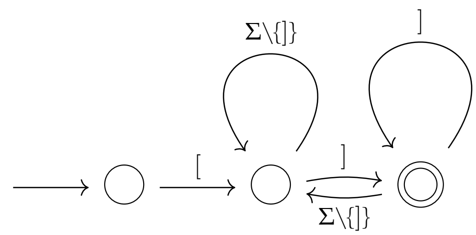
Problem 2.
Construct a deterministic finite automata (DFA) for the language in Problem 1.

Solution.



Problem 3.
Let Σ be the alphabet $\{0, 1, [,]\}$. Define L to be the set of strings over Σ that start with [and end with]. Notice that we allow extra [and] showing up within the string. For example, [10][01] and []] are valid strings in L, while [010]00 and][] are not in L. Design an NFA to recognize language L.

Solution.



Problem 4.
Let L be an automatic language. Construct an NFA that recognize the following language.

$$L' := \{w \in \Sigma^\star : \text{ no proper prefix of } w \text{ is in } L\}$$

(If we use the language L from Question 1, then L' is the set of strings that start with [and end with the first]. Your construction should work for general L.)

Solution.

Let D be a DFA that recognize the automatic language L, described by the tuple $(Q, s, A, \Sigma, \delta)$. We now construct a new ε-DFA N that recognizes the language $\text{noprefix}(L)$ as follows:

- $Q_N = Q$.
- Starting State: $S' = \{s\}$
- $A_N = A$.
- For every transition $\delta_D(q, a) = q'$ in D we add a transition $\delta_N(q', a) = q$ in N. And we remove every transition from p to other states q (even if $p = q$) if p is an accepting state in D. That is,

  $$\delta(p, a) = \emptyset$$

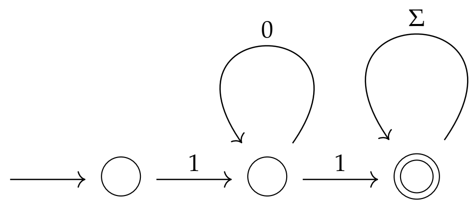
  for every $p \in A$ and $a \in \Sigma$.

Problem 5.
Define

$$L^{\geq} := \{1^k y : y \in \{0,1\}^\star \text{ and } y \text{ contains at least } k \text{ 1s for any } k \geq 1\}$$

Draw a DFA recognizing $L^{\geq}$ with minimum number of states, and prove that it is indeed minimum by constructing a fooling set of same number of prefixes. (No need to provide distinguishing suffixes that certify the fooling set.)

Solution.



Define a fooling set $F := \{\varepsilon, 0, 1\}$ with the distinguishing suffix 1.

Problem 6.
Define

$$L^{\leq} := \{1^k y : y \in \{0,1\}^\star \text{ and } y \text{ contains at most } k \text{ 1s for any } k \geq 1\}$$

Prove that $L^{\leq}$ is not regular by constructing a fooling set of infinite size. (You need to prove that it is indeed a fooling set.)

Solution.

Define the fooling set as $F := \{1^m 0 \mid m \in \mathbb{N}\}$. Then for two arbitrary prefixes $1^i 0$, $1^j 0 \in F$ with $i < j$, let the distinguishing suffix be 1^j. Then we have,

- $1^i 0 1^j \notin L$, because i is smaller than j and hence y has more than i 1s.
- $1^j 0 1^j \in L$, because y has j 1s.

This implies that F is a fooling set of infinite size, and thus L is not regular.

Problem 7.
Describe a Turing machine that achieves the following task.

- On input 1^n, leave $\#^{2^n}$ somewhere on the tape.

Solution.

Problem 8.
Argue that the following Turing machine model is equivalent in power to the standard Turing machine with a single tape:

- A Turing machine that never moves its cursor to the left 3 times in a row.

Solution. (q, 1)